

# ■ UZHAVU KAAPPAAN (■■■■■ ■■■■■■■■■■)

Smart Crop Rotation & Farmer Soil Feeding Action Plan  
Data Mode: ■ Verified Lab Ingestion (Device: soil-scout-03) · Certified P025 Model

Farmer: Gurpreet Singh | Land Area: 8.5 Acres  
Location: Ludhiana Granary (30.9010° N, 75.8573° E) | Irrigation: Canal Flood  
Plan Ref ID: #UK-P025-FARM-103 | Report Date: 03 Oct 2026, 12:02 AM IST  
Live Reading: Active & Calibrated

## 1. Live Sensor Diagnostic & Soil Fertility Telemetry

| ■■ Sunlight / Light        | ■ Soil Moisture             | ■■ Temperature               | ■ TDS Minerals                  | ■ Probe Reliability                                     |
|----------------------------|-----------------------------|------------------------------|---------------------------------|---------------------------------------------------------|
| 74%<br>Optimal photoperiod | 45%<br>Rhizosphere moisture | 31.5 °C<br>Soil probe sensor | 420 ppm<br>Conductivity & salts | ■ Reliable<br>Soil Scout v2                             |
| Nutrient Parameter         | Real-Time Value             | Target Range                 | Deficit / Status                | Agronomic Impact & Crop Response                        |
| Nitrogen (N)               | 50.0 kg/ha                  | 80 - 160 kg/ha               | -30.0 kg/ha (Deficit)           | Restricts leaf canopy; urgent pulse rotation needed     |
| Phosphorus (P)             | 42.0 kg/ha                  | 30 - 60 kg/ha                | Sufficient                      | Affects root elongation & early plant establishment     |
| Potassium (K)              | 48.0 kg/ha                  | 60 - 120 kg/ha               | -12.0 kg/ha (Sub-optimal)       | Adequate for disease resistance, cell turgor & pod fill |
| Soil pH                    | 7.80                        | 6.0 - 7.5                    | Alkaline (Gypsum required)      | Governs cation exchange & micronutrient availability    |
| Organic Carbon (OC)        | 0.45%                       | 0.80 - 1.50%                 | -0.35% Deficit                  | Governs moisture holding capacity & soil microbiome     |

## 2. Prescribed Soil Feeding & Fertilizer Dosage Schedule (for 8.5 Acres Land)

| Feeding Stage                        | Recommended Inputs                                                | Dose / Acre                         | Total (8.5 Ac)                     | Application Method & Agronomic Objective                                                              |
|--------------------------------------|-------------------------------------------------------------------|-------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------|
| Basal Land Prep<br>(Prior to Sowing) | FYM / Vermicompost<br>Neem Cake<br>Bio-fertilizer (Rhizobium+PSB) | 4.0 Tonnes<br>100 kg<br>2 kg + 2 kg | 34.0 Tonnes<br>850 kg<br>34.0 kg   | Incorporate during final ploughing to boost Organic Carbon & inoculate root nodules                   |
| Sowing Time<br>(Basal Starter)       | DAP (Di-Ammonium Phos.)<br>MOP (Potash)<br>Zinc Sulphate (ZnSO4)  | 25 kg<br>15 kg<br>10 kg             | 212.5 kg<br>127.5 kg<br>85.0 kg    | Apply in bands 5cm below seed line for rapid root growth & zinc deficiency correction                 |
| Vegetative Growth<br>(25-30 Days)    | Urea (Top Dressing)                                               | 15 kg (-40% saved)                  | 127.5 kg<br>Save 85 kg (Rs. 2,975) | Wheat biologically fixes nitrogen; saves 85 kg urea (Rs. 2,975) across 8.5 acres and prevents lodging |
| Flowering & Pods<br>(45-50 Days)     | 19:19:19 Soluble NPK<br>Borax (0.2% Boron)                        | 1.0 kg (Foliar)<br>200 g (Foliar)   | 8.5 kg<br>1.70 kg                  | Early morning foliar spray to prevent flower drop and ensure full pod filling                         |

## 3. Agronomic Risk Assessment & Recommended Restorative Crop Rotation

| Season 1 (Immediate Restoration)                                                                                                                                                                                                                                                                                     | Season 2 (Soil Building)                                                                                                                                                                                                                                                               | Season 3 (Biomass & Stabilization)                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Wheat (Top Match: 88.5%)</b> <ul style="list-style-type: none"><li>• Crop Family: Cereal</li><li>• Duration: 100-120 Days</li><li>• Est. Net Profit: Rs. 29,000 / acre</li><li>• Biological Role: Deep fibrous root system absorbs subsoil nutrients, adds carbon-rich stubble, and completes recovery.</li></ul> | <b>Green Gram</b> <ul style="list-style-type: none"><li>• Crop Family: Legume</li><li>• Duration: 65-85 Days</li><li>• Est. Net Profit: Rs. 6,482 / acre</li><li>• Biological Role: Biologically fixes 35-45 kg N/ha naturally, cuts synthetic urea, and breaks pest cycles.</li></ul> | <b>Rice</b> <ul style="list-style-type: none"><li>• Crop Family: Cereal</li><li>• Duration: 100-120 Days</li><li>• Est. Net Profit: Rs. 15,000 / acre</li><li>• Biological Role: Deep fibrous root system absorbs subsoil nutrients, adds carbon-rich stubble, and completes recovery.</li></ul> |

## 4. Multi-Season Financial Returns & Soil Health Recovery Trajectory

| Rotation Stage    | Crop Cultivated      | Cost / Ac  | Gross / Ac | Net Profit / Ac | Farm Total (8.5 Ac) | Soil Trajectory     |
|-------------------|----------------------|------------|------------|-----------------|---------------------|---------------------|
| Baseline          | Depleted State       | Rs. 36,000 | Rs. 48,000 | Rs. 12,000      | Rs. 102,000         | 61 / 100 (Current)  |
| Season 1 (Kharif) | Wheat                | Rs. 18,000 | Rs. 15,442 | Rs. 29,000      | Rs. 246,500         | 68 / 100 (+7 pts)   |
| Season 2 (Rabi)   | Green Gram           | Rs. 11,000 | Rs. 17,482 | Rs. 6,482       | Rs. 55,097          | 75 / 100 (+7 pts)   |
| Season 3 (Zaid)   | Rice                 | Rs. 24,000 | Rs. 39,000 | Rs. 15,000      | Rs. 127,500         | 82 / 100 (Restored) |
| 3-Season Total    | Restorative Rotation | Rs. 53,000 | Rs. 71,924 | Rs. 50,482 / ac | Rs. 429,097 Total   | 82 / 100 (Restored) |

- **Mandatory Agronomic Field Guidelines for Gurpreet Singh:**
1. **Bio-Inoculation:** Treat Wheat seeds with *Rhizobium* bio-fertilizer @ 25g/kg seed to maximize root nodulation.
  2. **Input Cost Savings:** Restricting top-dressed urea to 15 kg/acre saves 85 kg chemical urea (Rs. 2,975 direct savings across 8.5 acres) thanks to legume atmospheric N fixation.
  3. **Crop Residue Retention:** Do not burn crop stubbles. Plough haulms back into soil to lift Organic Carbon from 0.45% towards 0.80%.
  4. **Precision IoT Monitoring:** Keep Soil Scout probe clean; observe moisture reading (45%) to trigger irrigation when soil reaches 35%.

Certified by: UZHAVU KAAPPAAN P025 Model  
Telemetry: Live Soil Scout Probe #soil-scout-03 · 45,000+ Agricultural Training Records

Authorized Agronomist Extension Officer:  
Department of Agriculture & Precision Agronomy, Ludhiana Granary